JOHAN DAVID RODRIGUEZ CASTRO

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PROFESSIONAL PROFILE

Final-semester Systems and Software Engineering student (UPB Bucaramanga; expected to graduate in 2026) working at the intersection of applied machine learning and security engineering. I ship tools rather than findings: an open-source supply-chain scanner, a phishing classifier that audited its own benchmark and reported the honest number instead of the flattering one, and a vulnerability prioritisation dashboard that measured away the advantage it was built to show. Verified C2 English (EF SET).

TECHNICAL SKILLS

Data, ML & AI: Model training and evaluation; LightGBM, XGBoost, TensorFlow / Keras, scikit-learn; data-leakage detection and honest evaluation design; SHAP explainability; grouped, temporal and walk-forward validation; probability calibration; AI agents (smolagents, CrewAI) and MCP; FinBERT NLP; pandas / NumPy

Security & Detection: Software supply-chain security (slopsquatting), detection-as-code, MITRE ATT&CK mapping, risk-based vulnerability prioritisation (CISA KEV / EPSS / CVSS), phishing and URL analysis, TOTP 2FA, secure SDLC and CI security gates

Cloud & DevOps: AWS (IAM, VPC, EC2, Auto Scaling, ALB, CloudWatch), Terraform (IaC), Docker, GitHub Actions CI/CD, k6, Google Cloud (BigQuery ML, Cloud Storage, DLP)

Programming: Python, TypeScript / JavaScript, SQL, Dart, HCL, LaTeX

Databases: PostgreSQL, MySQL, SQLite, DuckDB, ChromaDB; hybrid SQL / NoSQL architectures

Methods & Tools: Agile / Scrum, spec-driven development, pytest, Playwright, mypy strict, UML modelling, Git

EXPERIENCE

Developer / Technical Support (Volunteer) GOATGuard client: github.com/Yoyagm/goatguard-app | Jan 2026 – Present | Pontificia Universidad Bolivariana

- Technical support for the institution's end users.
- Endpoint threat detection: identifying and analysing port scans and network intrusion attempts.
- Automatic classification of network assets (routers, printers, cameras) with a real-time monitoring dashboard and KPIs.
- GOATGuard Android client (Flutter), a team network-monitoring project: dashboard, inventory, alerts, TOTP 2FA. Author of 19 of the client repository's 25 commits.

KEY TECHNICAL PROJECTS

SlopGuard — *Open-Source Supply-Chain Security Scanner* github.com/Yoyagm/slopguard | 2026 | Independent

- Built a pre-install guard that scores PyPI and npm dependencies against hallucinated, typosquatted and malicious names *before* any code executes, using a five-layer detection engine with zero runtime dependencies.
- Enforced correctness with a CI gate at ≥90 % global coverage (≥95 % on critical paths) and eight import-linter architecture contracts, and shipped four integration points from one core: CLI, pre-commit hook, GitHub Action and self-hostable SaaS.

Phishing URL Detector — *Explainable Detection & Leakage Audit* yoyagm.github.io/phishing-url-detector | 2026 | Independent

- Audited PhiUSIIL (UCI id 967, 235,795 URLs) before training and measured two independent label leaks: a one-line rule scores 99.67 % accuracy, and hand-rolled lexical features on the raw URL — the obvious fix — still score 99.51 % while measuring nothing.
- Retrained on canonicalised host names only and published the honest result: ROC-AUC 0.937 and 70.5 % recall at a 1 % false-positive rate, against phishing submitted two years after the training data was collected.
- Shipped it: exact TreeSHAP reimplemented in dependency-free JavaScript (parity-tested at 1e-12 against the Python reference), 113 tests including leakage guards with negative controls, and a containerised FastAPI service.

Vulnerability Prioritisation Dashboard — *CISA KEV + EPSS + NVD* github.com/Yoyagm/vuln-prioritization-dashboard | 2026 | Independent

- Built a reproducible ingestion pipeline over three public feeds ranking 359,399 live CVEs, with committed data contracts and schema tests that fail the build when a feed changes shape.
- Reproduced the industry claim that EPSS is 6.1× more efficient than CVSS, then showed with frozen scores over five cutoffs that most of it is evaluation artefact: EPSS wins the head of the queue (1.2–1.8×) and loses the long tail (0.5–0.7×).

Also: AWS three-tier HA infrastructure in Terraform + GitHub Actions (eval-final-dgiti) | autonomous ML and multi-agent market pipeline, 505 tests. Case studies: johanrodriguez.is-a.dev/projects.

EDUCATION

B.S. in Systems and Software Engineering Feb 2023 – Present | Pontificia Universidad Bolivariana

- Final semester. Coursework in software architecture and system design; AWS Academy Cloud Foundations labs 1–6 (IAM, VPC, EC2, EBS, Auto Scaling, Load Balancing).
- Scrum Master of an academic high-availability infrastructure project (Jun–Nov 2025): ran the Scrum cycle and contributed to the design of a system sized for 100,000 concurrent users with a sub-500 ms latency objective (design targets, not measured outcomes).

Postgraduate Specialisation in Information Security Sep 2026 – Nov 2026 (in progress) | Pontificia Universidad Bolivariana

CERTIFICATIONS

Google Cloud (2026) — 7 skill badges: Secure Lakehouse Data • Protect Sensitive Data with Data Loss Prevention • Create ML Models with BigQuery ML • Use APIs to Work with Cloud Storage • Gemini Enterprise (multi-agent workflows, first application) • Vibe Coding and MCP. **Google** (2026) — DeepMind: Train A Small Language Model.
Cisco Networking Academy (2026) — Introduction to Cybersecurity. **Hugging Face** (2026) — AI Agents Course. **freeCodeCamp** (2026) — Machine Learning with Python • Data Analysis with Python • Scientific Computing with Python • Data Visualization • JavaScript Algorithms and Data Structures.
Languages: Spanish — Native • English — C2 Proficient (EF SET 74/100, verified). All credentials publicly verifiable at linkedin.com/in/johan-rodriguez-97bb29323.